D4.2 Stability and change

D4 Continuity and change
Ecosystems

SL

Understandings:

- D4.2.1—Stability as a property of natural ecosystems
- D4.2.2—Requirements for stability in ecosystems
- D4.2.3—Deforestation of Amazon rainforest as an example of a possible tipping point in ecosystem stability
- D4.2.4—Use of a model to investigate the effect of variables on ecosystem stability
- D4.2.5—Role of keystone species in the stability of ecosystems
- D4.2.6—Assessing sustainability of resource harvesting from natural ecosystems
- D4.2.7—Factors affecting the sustainability of agriculture
- D4.2.8—Eutrophication of aquatic and marine ecosystems due to leaching
- D4.2.9—Biomagnification of pollutants in natural ecosystems
- D4.2.10—Effects of microplastic and macroplastic pollution of the oceans
- D4.2.11—Restoration of natural processes in ecosystems by rewilding

Key words:

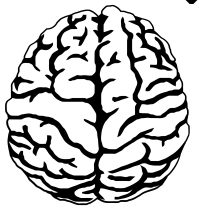
Deforestation
Tipping point
Percentage change
Mesocosm
Keystone species
Sustainable
Harvesting
Agriculture
Mechanised
Tilling
Monocrops
Fertilisers
Insecticides
Permaculture
Leeching
Eutrophication
Biochemical oxygen demand (BOD)
Biomagnification
Bioaccumulation
DDT
Microplastics
Mercury
Rewilding

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.



Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.



Guiding questions

- *What features of ecosystems allow stability over unlimited time periods?*
- *What changes caused by humans threaten the stability of ecosystems?*

D4.2.1—Stability as a property of natural ecosystems



If left with consistent environmental conditions, an ecosystem will find a “natural balance”. This, in theory, could lead to a stable ecosystem that does not undergo any major changes.

What type of ecosystem do you think is most likely to persist unchanged for long periods of time?

How old are the oldest ecosystems found on Earth?

Rainforests

Amazon rainforest - 55 million years old

Taman Negara - 130 million years old

Borneo lowland rainforest - 140 million years old

Daintree rainforest - 180 million years old.

Can you locate the named rainforests on this map?





What conditions do these rainforests share?

Why is this important?

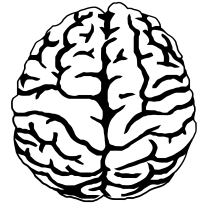
Rainforest

- Consistent high rainfall
- Consistent sunlight

Lots of photosynthesis!



D4.2.2—Requirements for stability in ecosystems



What biotic and abiotic conditions are required for a stable ecosystem?

What biotic and abiotic conditions are required for a stable ecosystem?

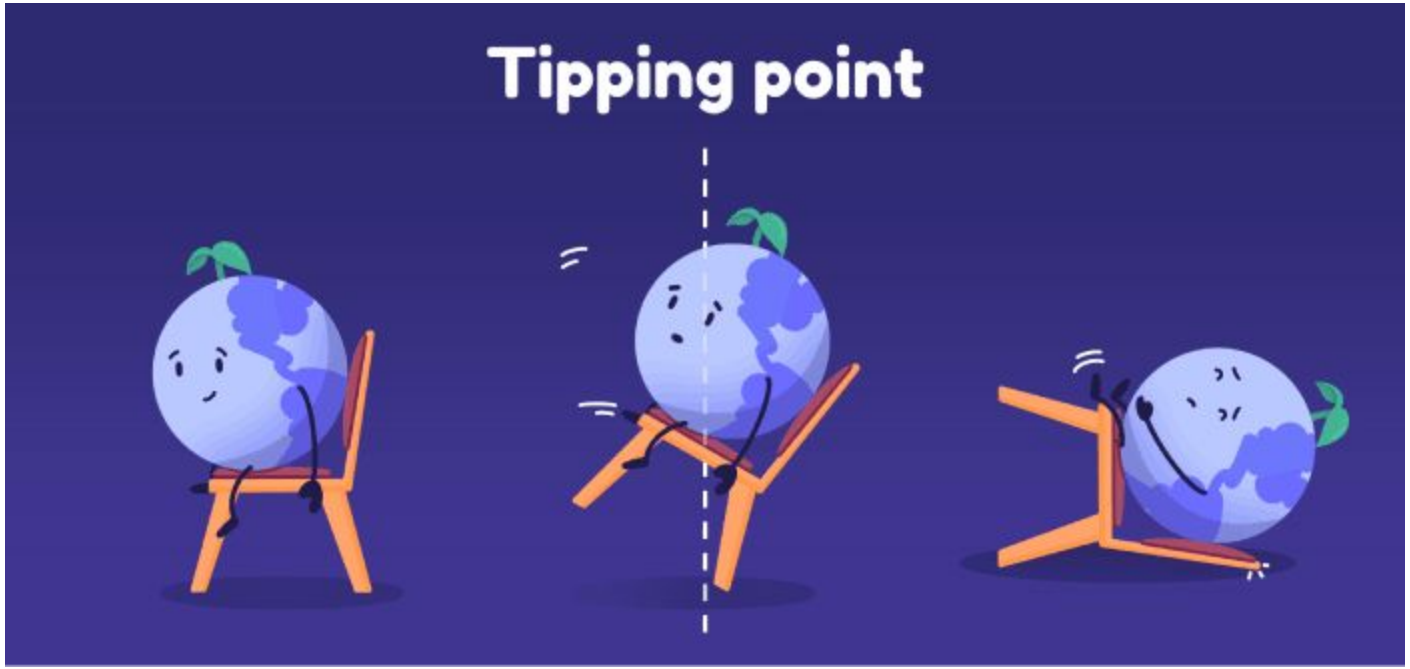
High level and consistent supply of energy (usually as sunlight)

Robust recycling of nutrients (healthy decomposers)

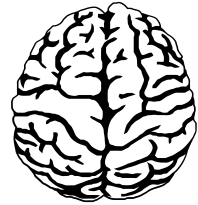
Genetic diversity (resilient to change)

Stable climate (within tolerance levels)

D4.2.3—Deforestation of Amazon rainforest as an example of a possible tipping point in ecosystem stability


Tipping point



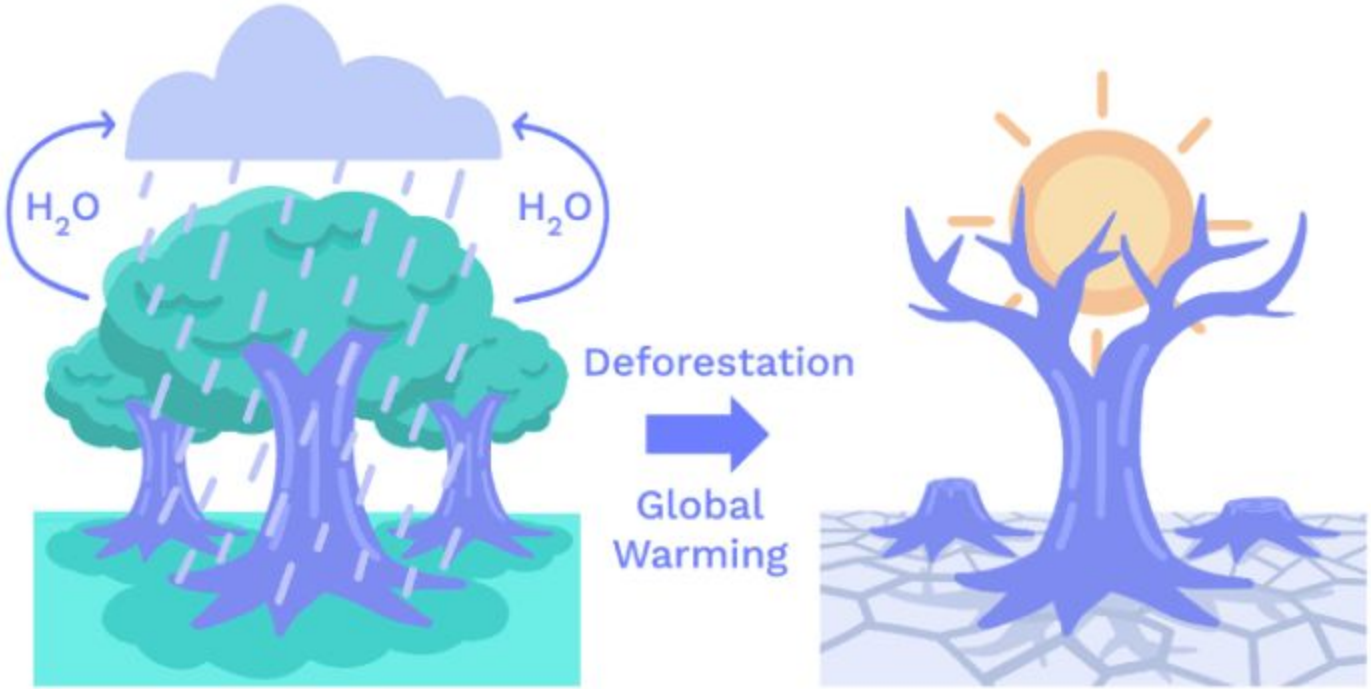


Why is there so much rain in the amazon rainforest?

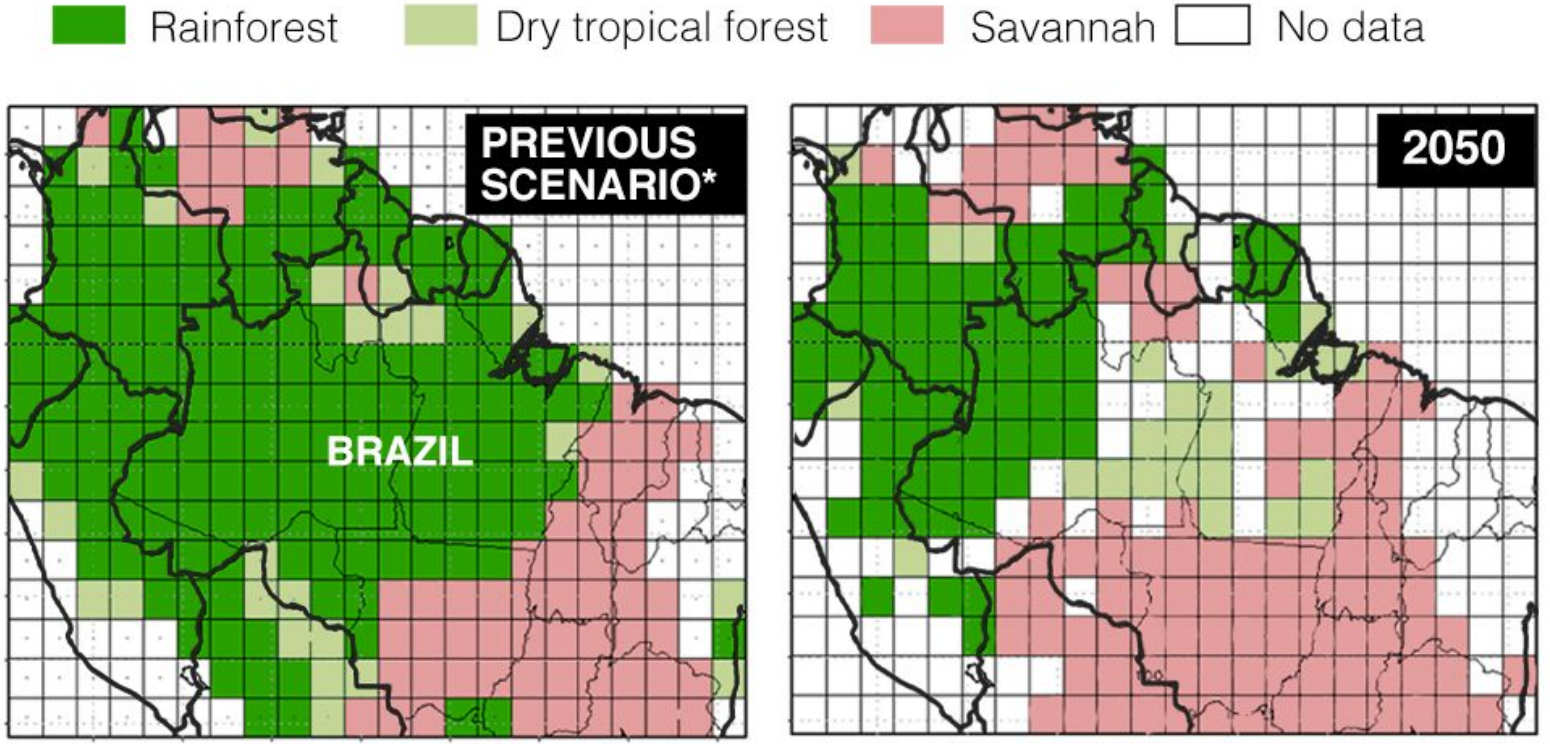
Why is there so much rain in the amazon rainforest?

Evapotranspiration!

Amazon Rainforest Tipping Point



How will the Amazon look after 'tipping point'?

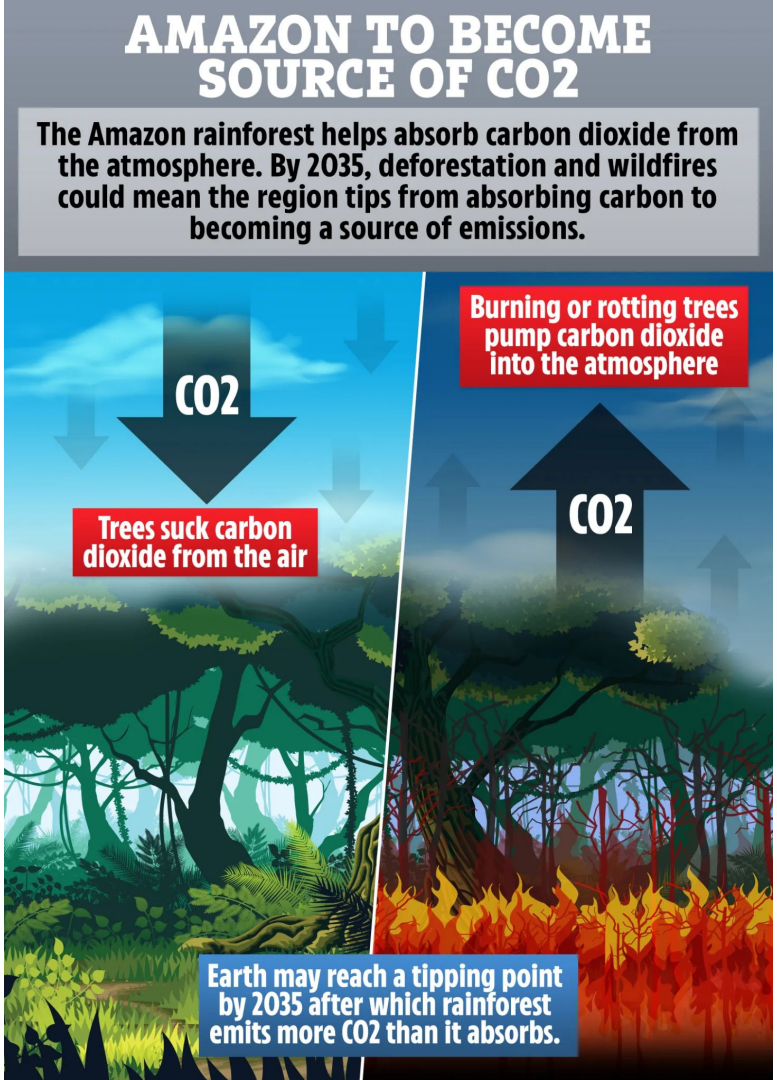


*Relative to the average climate of 20th century

Source: Carlos Nobre, Thomas Lovejoy and Gilvan Sampaio, 2016 PNAS

BBC

In some parts of Brazil the Amazon rainforest is already a net carbon source.



Uncertainty exists over how much rainforest needs to be lost to reach this tipping point.

Some suggest a loss of 25%.

It is estimated that we are already at 20% loss from deforestation.

Percentage change

What is the percentage change in amazon rainforest cover between 1970 and 2018?



$$\text{Percentage change} = \frac{\text{Change}}{\text{Original}} \times 100$$

D4.2.4—Use of a model to investigate the effect of variables on ecosystem stability

Remember: “Discuss” requires pros and cons!

Mesocosm

Discuss why a controlled artificial environment (such as a mesocosm) might be more appropriate for studying an ecosystem?

Are there any downsides?



Mesocosm

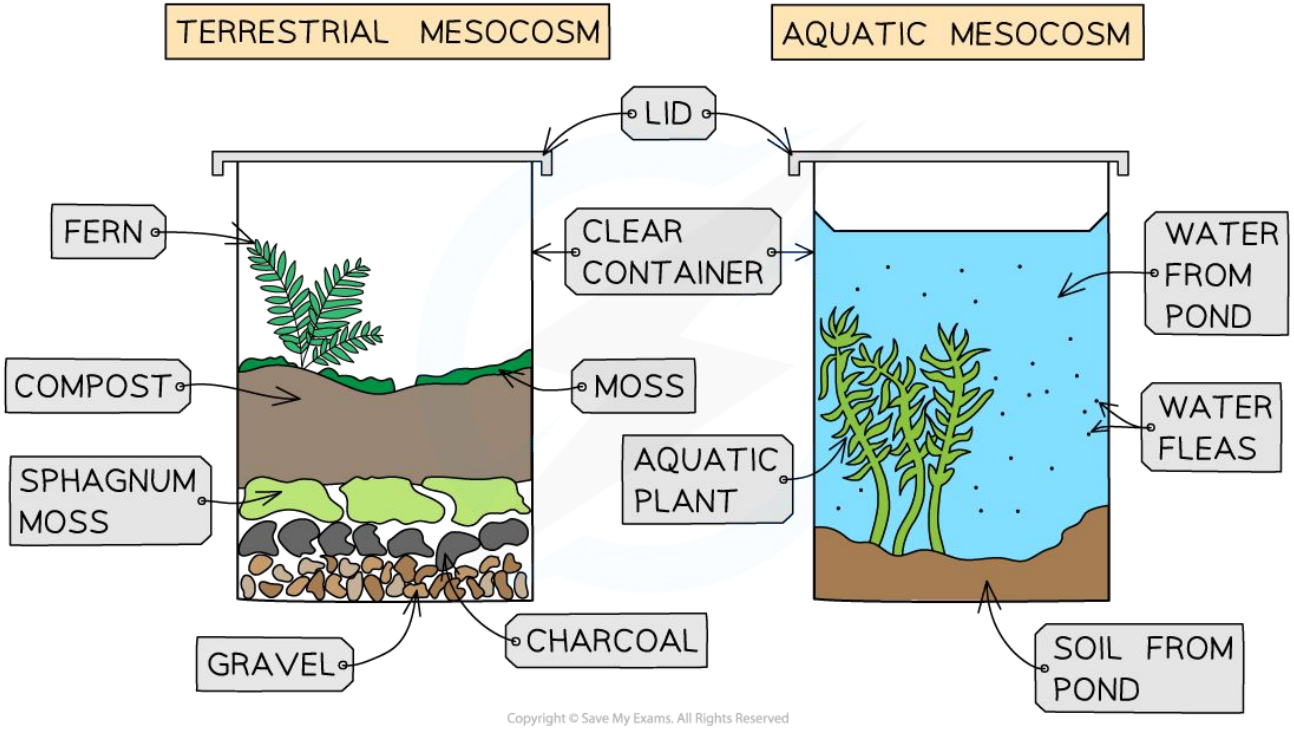
Discuss why a controlled artificial environment (such as a mesocosm) might be more appropriate for studying an ecosystem?

Are there any downsides?

- Able to manipulate variables
- Better able to control variables and establish causation
- Takes up less space
- Never able to fully recreate natural ecosystem
- Cannot be done with large animals
- Ethical issues



Mesocosm



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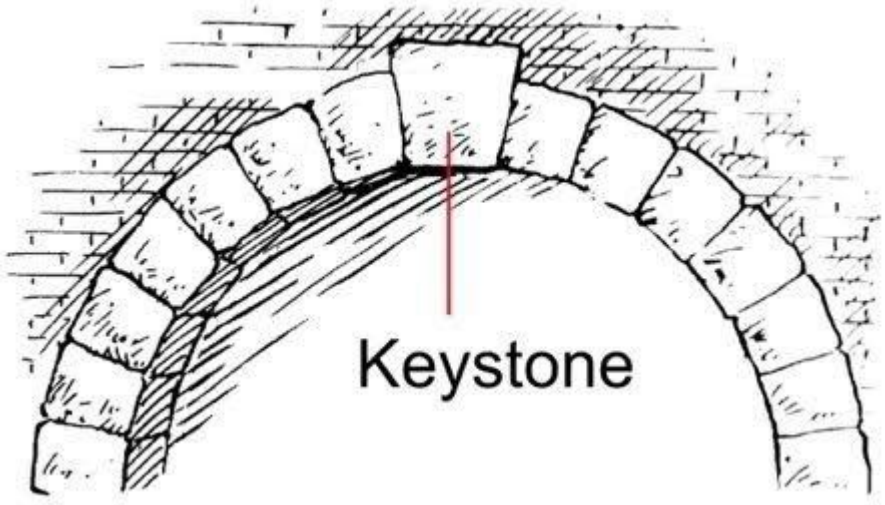
Nature of Science



Care and maintenance of the mesocosms should follow IB experimental guidelines.

D4.2.5—Role of keystone species in the stability of ecosystems

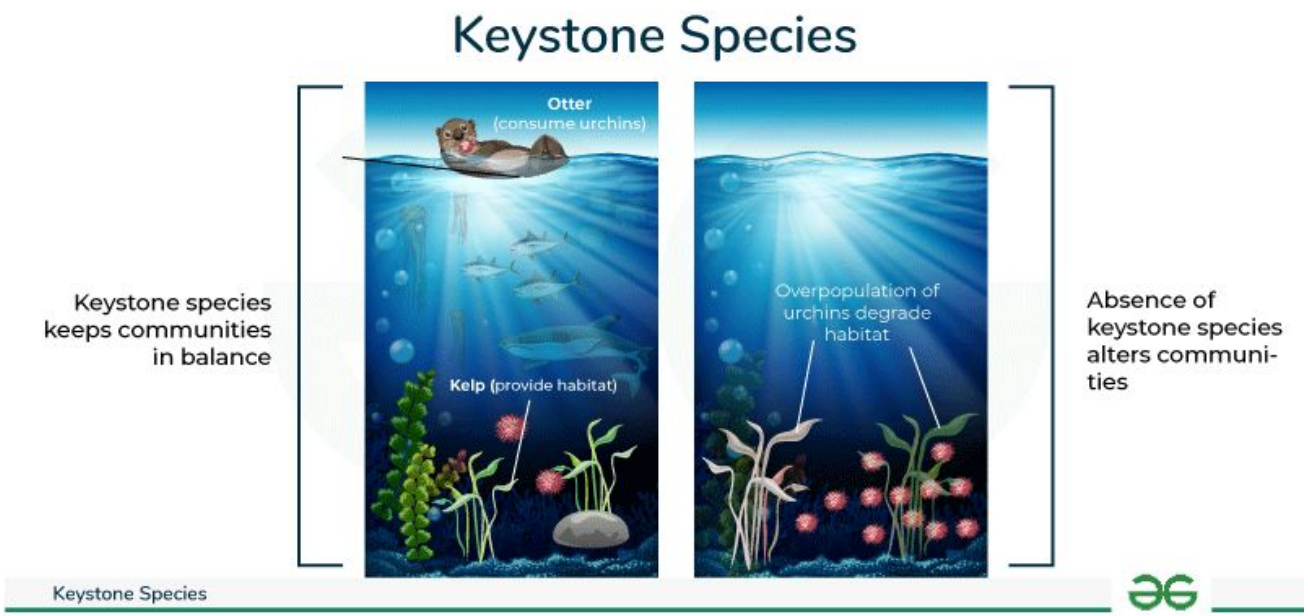
What happens if you remove the keystone?

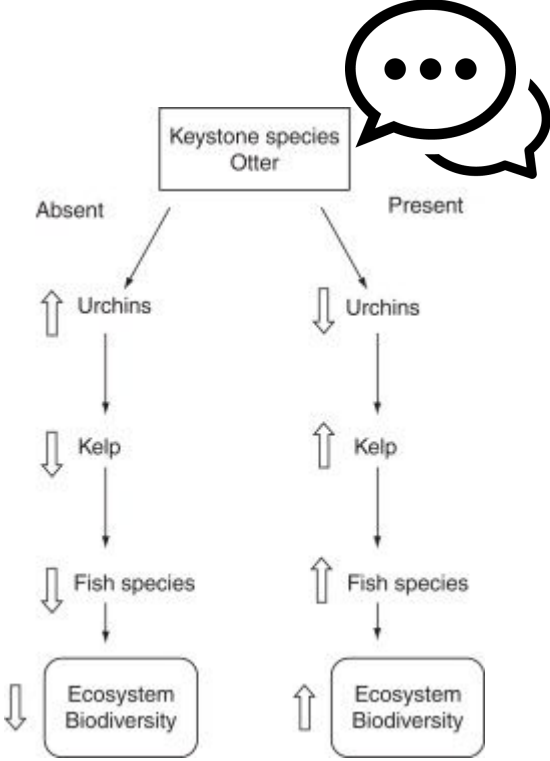


A keystone species is an organism that has a disproportionately large impact on its ecosystem. If removed, the ecosystem changes dramatically.

For example bees.







Using these two graphics, explain how the otter acts as a keystone species



Can you think of more examples of keystone species?

KEYSTONE SPECIES

WHAT ARE KEYSTONE SPECIES?

Keystone species are species that have a massive impact on an ecosystem, such that if their presence is removed, the ecosystem will dramatically change.

WHAT HAPPENS IF KEYSTONE SPECIES ARE REMOVED?

Their removal in an ecosystem can result in a drastic shift in population and overall stability of the ecosystem.

EXAMPLES OF KEYSTONE SPECIES


PURPLE SEA STAR


GRAY WOLVES


ELEPHANTS


MANGROVE TREES


BEAVERS


SHARKS


SAGUARO CACTUS


GRIZZLY BEARS


SEA OTTERS


BEES

OTHER ORGANISMS CRUCIAL TO ECOSYSTEMS

UMBRELLA SPECIES

- Their influence is tied to geographic range. Most species are migratory or have large territories they roam, influencing species across that region. A prime example is a SIBERIAN TIGER.

FOUNDATION SPECIES

- These species have a significant influence by creating or maintaining a habitat. CORALS are the best examples as coral reefs create and maintain habitats for thousands of species of marine life.

INDICATOR SPECIES

- They are indicators of a change in the ecosystem. Scientists use them as early warning signs if there is a change in the ecosystem. Prime example is an OYSTER



Outforia

D4.2.6—Assessing sustainability of resource harvesting from natural ecosystems



What is a key feature of a natural resource that can be harvested sustainably?

What is a key feature of a natural resource that can be harvested sustainably?

It reproduces at a faster rate than it is harvested!



Bamboo is a very sustainable wood

Very rapid growth

Minimal water and fertiliser requirements

Very strong






This Is A Sustainable Fish

Sardines are a small and plentiful fish species.

Locally caught or fish with the MSC label are more likely to be sustainably caught



CERTIFIED SUSTAINABLE SEAFOOD
MSC
www.msc.org
TM

D4.2.7—Factors affecting the sustainability of agriculture



Most food comes from large scale intensive farming

- Mechanised - large machinery
- Tilling - breaking up the soil
- Monocrops - efficient harvesting and commerce
- Chemical fertilisers - replace nutrients lost to plant growth
- Insecticides - protect crop

Explain how these methods can damage the environment and be considered unsustainable

Mechanised - Polluting use of fossil fuels

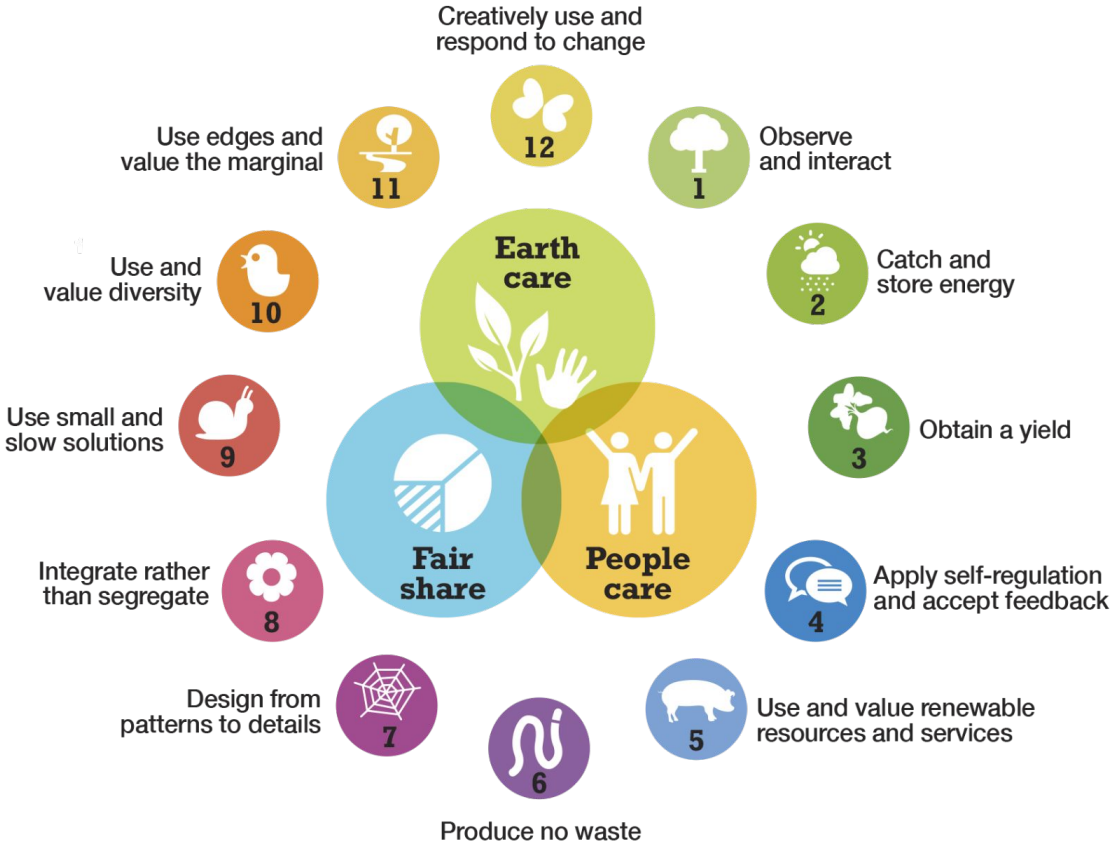
Tilling - Leads to soil erosion

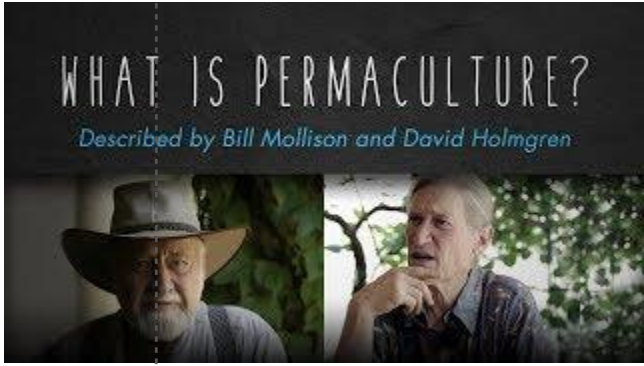
Monocrops - Encourage pests (leading to more insecticides)

Chemical fertilisers - Can lead to groundwater pollution and eutrophication

Insecticides - Fears over effects on stability of ecosystem. Health concerns

PERMACULTURE PRINCIPLES

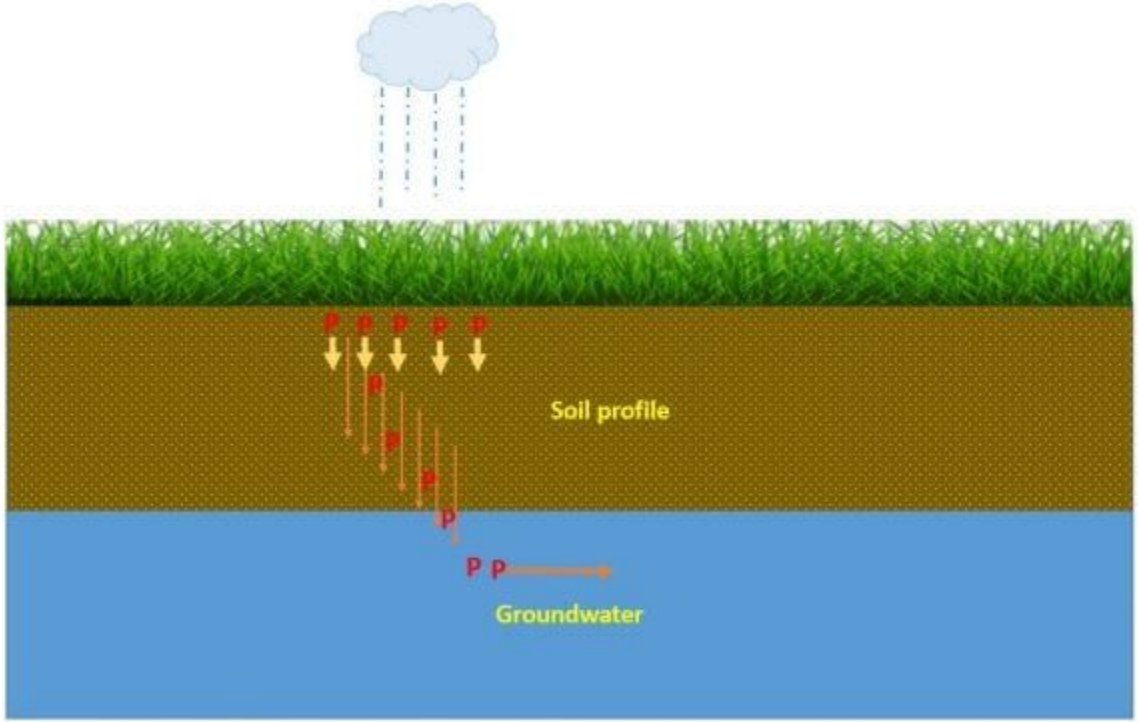


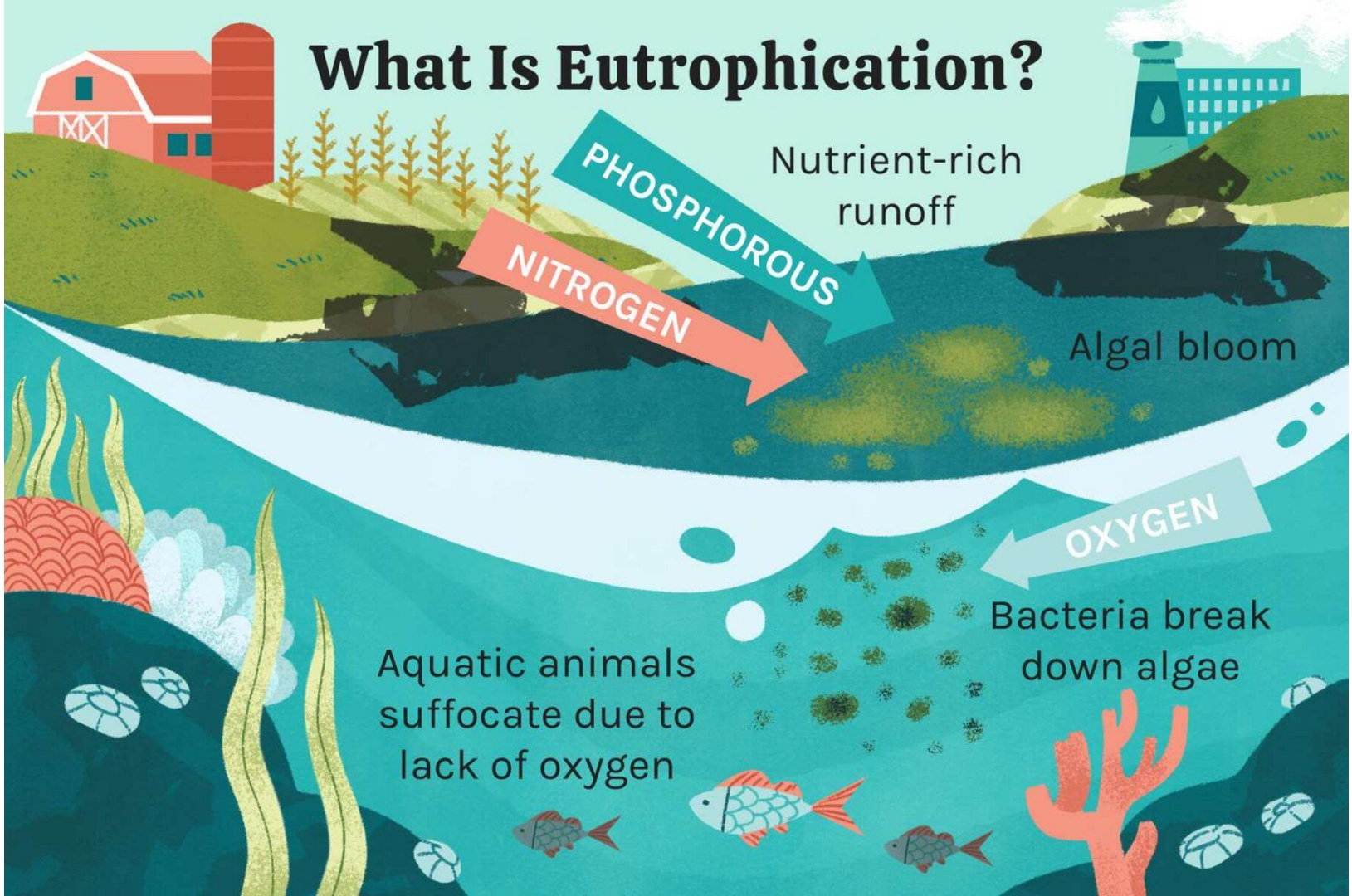


Do not need to know these!

D4.2.8—Eutrophication of aquatic and marine ecosystems due to leaching

Leaching




What Is Eutrophication?
PHOSPHOROUS
NITROGEN
Nutrient-rich runoff
Algal bloom
OXYGEN
Bacteria break down algae
Aquatic animals suffocate due to lack of oxygen

Eutrophication

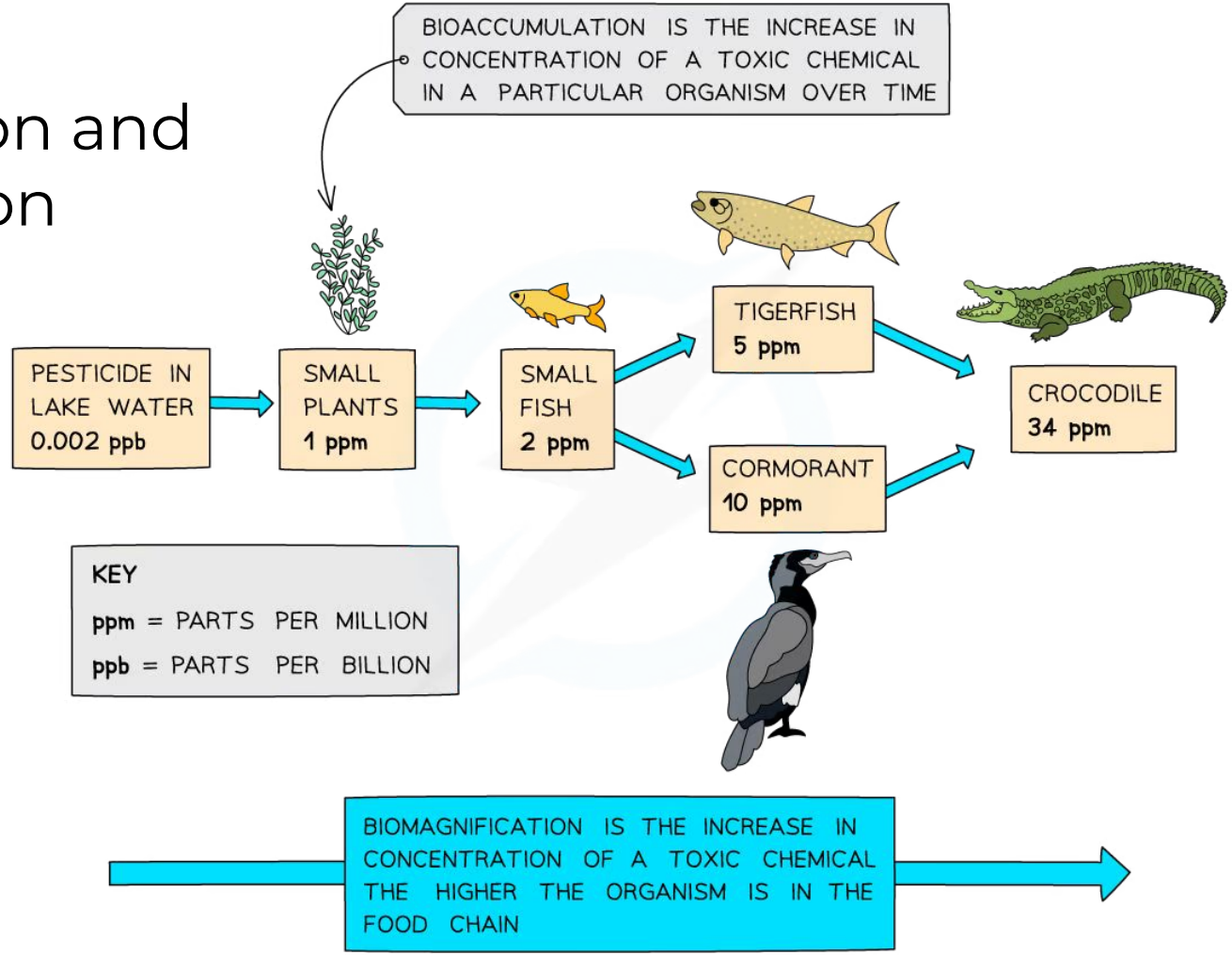
- Nutrients (often from excessive fertiliser, particularly nitrogen and phosphate) leaches into waterways
- Explosion of growth of algae on surface
- Sunlight not able to reach aquatic plants
- Plants and algae die
- Bacteria feed on dead plants and algae
- Bacteria use up oxygen in water (increase biochemical oxygen demand, **BOD**)
- Lack of oxygen in water causes death of organisms living in water

D4.2.9—Biomagnification of pollutants in natural ecosystems

Bioaccumulation and Biomagnification

When an organism consumes another organism it can process and utilise the organic materials but the toxins can not be used and so accumulate.

(Organisms eat larger quantities of the previous trophic level)



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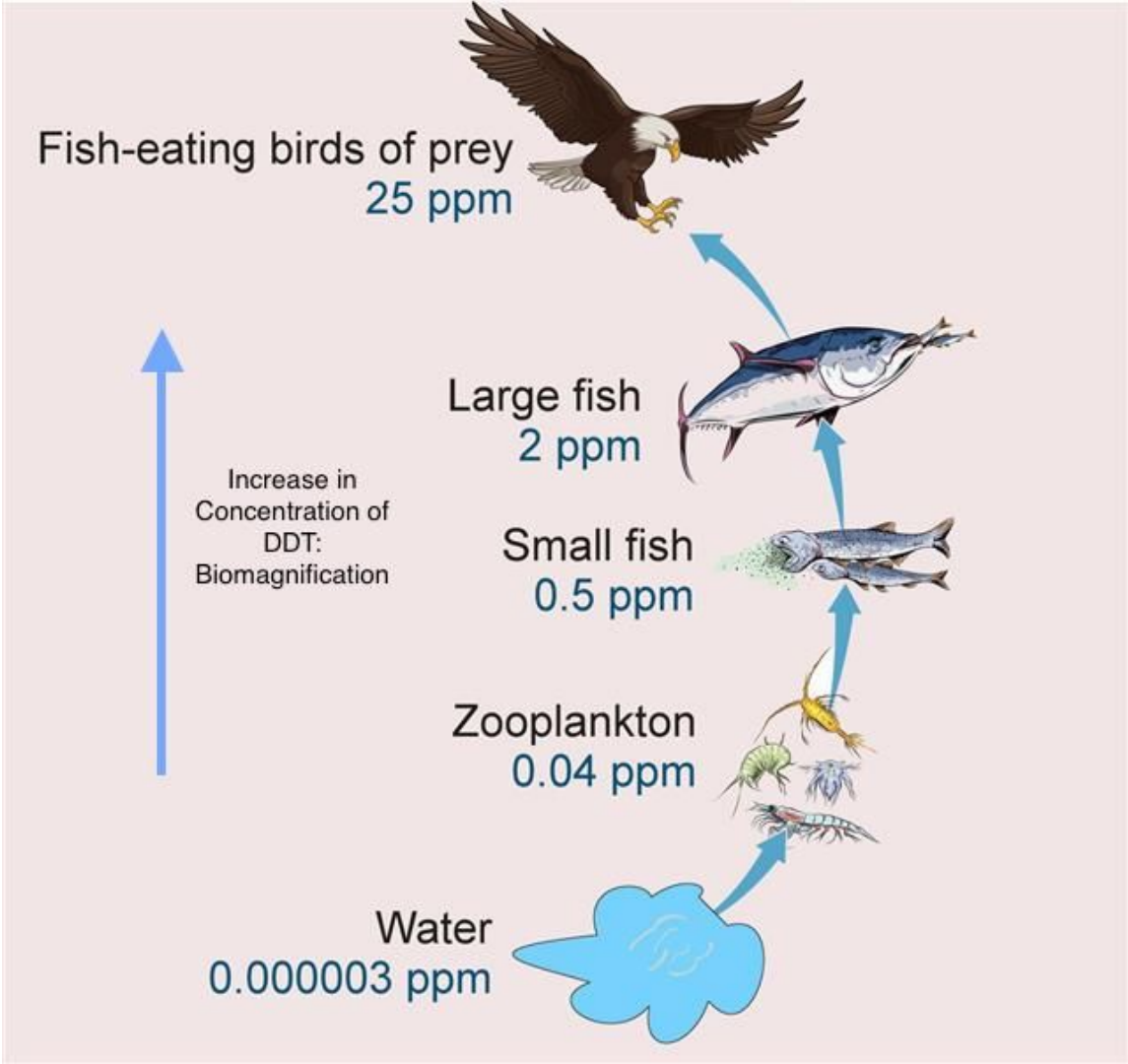
DDT

DDT used extensively from the 1940s as a pest control to prevent diseases like malaria.

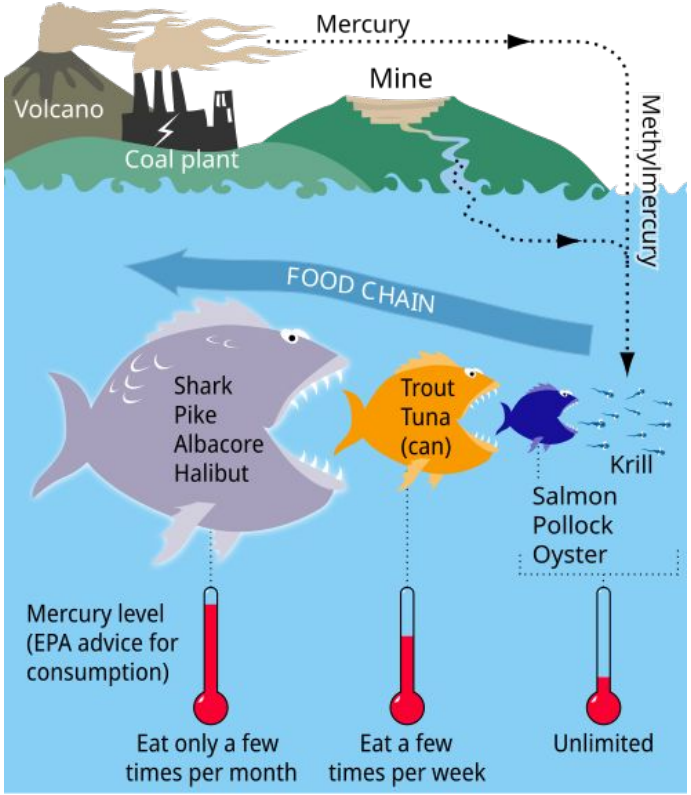
Accumulated in fatty tissue.

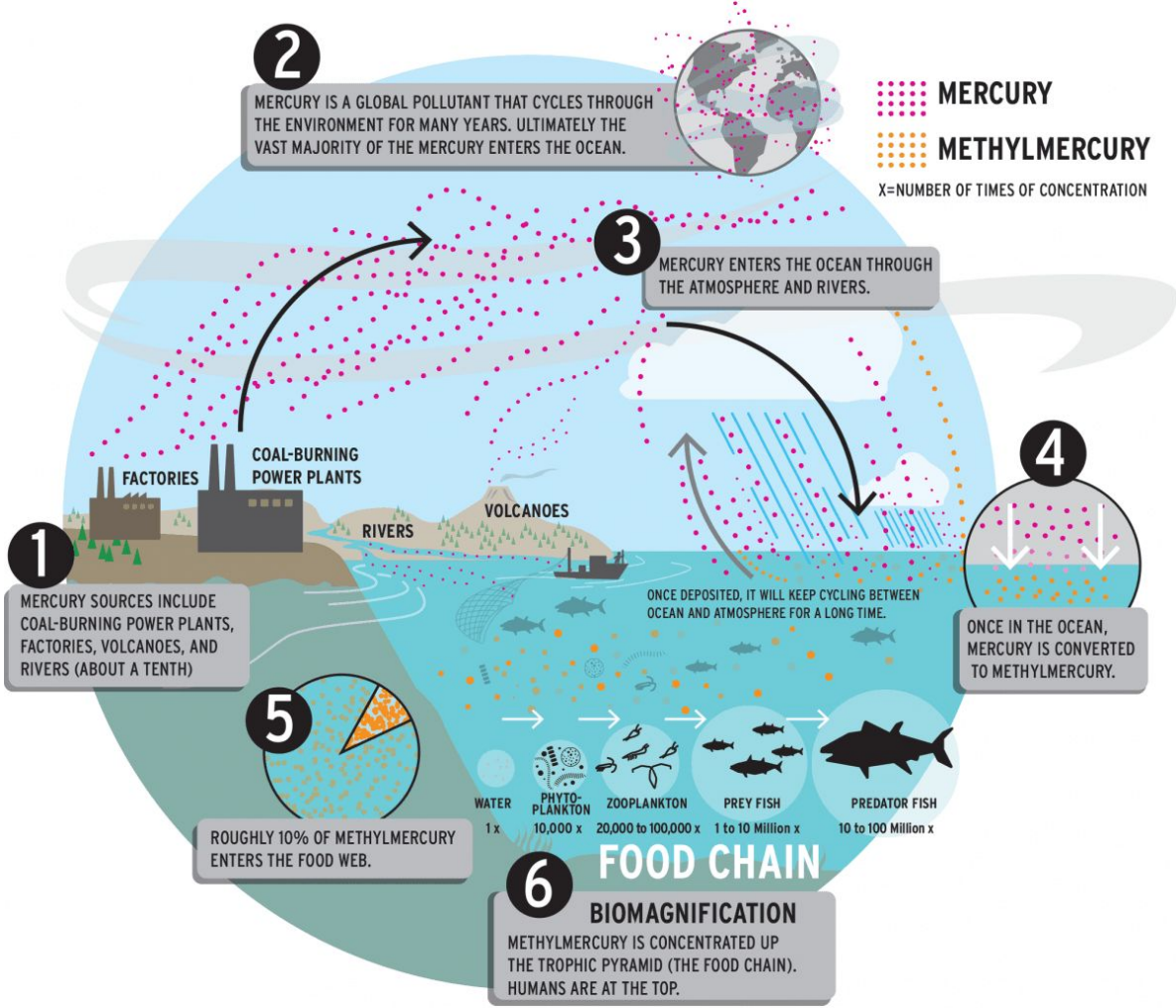
Led to egg shell thinning. Carcinogenic.

Banned in 1978 in USA, still used in some countries today.



Mercury


Mercury
Volcano
Mine
Coal plant
Methylmercury
FOOD CHAIN
Shark
Pike
Albacore
Halibut
Trout
Tuna
(can)
Krill
Salmon
Pollock
Oyster
Mercury level
(EPA advice for
consumption)
Eat only a few
times per month
Eat a few
times per week
Unlimited


2
MERCURY IS A GLOBAL POLLUTANT THAT CYCLES THROUGH THE ENVIRONMENT FOR MANY YEARS. ULTIMATELY THE VAST MAJORITY OF THE MERCURY ENTERS THE OCEAN.
MERCURY
METHYLMERCURY
X=NUMBER OF TIMES OF CONCENTRATION
3
MERCURY ENTERS THE OCEAN THROUGH THE ATMOSPHERE AND RIVERS.
FACTORIES
COAL-BURNING POWER PLANTS
RIVERS
VOLCANOES
1
MERCURY SOURCES INCLUDE COAL-BURNING POWER PLANTS, FACTORIES, VOLCANOES, AND RIVERS (ABOUT A TENTH)
ONCE DEPOSITED, IT WILL KEEP CYCLING BETWEEN OCEAN AND ATMOSPHERE FOR A LONG TIME.
4
ONCE IN THE OCEAN, MERCURY IS CONVERTED TO METHYLMERCURY.
5
ROUGHLY 10% OF METHYLMERCURY ENTERS THE FOOD WEB.
WATER
1 x
PHYTO-PLANKTON
10,000 x
ZOOPLANKTON
20,000 to 100,000 x
PREY FISH
1 to 10 Million x
PREDATOR FISH
10 to 100 Million x
FOOD CHAIN
6
BIOMAGNIFICATION
METHYLMERCURY IS CONCENTRATED UP THE TROPHIC PYRAMID (THE FOOD CHAIN). HUMANS ARE AT THE TOP.

D4.2.10—Effects of microplastic and macroplastic pollution of the oceans

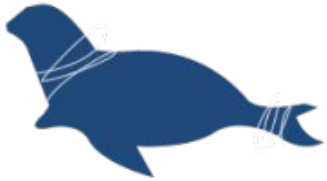


IMPACTS OF MARINE DEBRIS



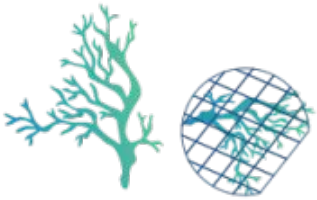
INGESTION

Animals mistakenly eat plastic and other debris.



ENTANGLEMENT & GHOSTFISHING

Marine life gets caught and killed in ghost nets, trapped in derelict gear, and entangled in plastic bands and other marine debris.



HABITAT DAMAGE

Heavy marine debris crushes sensitive habitat, such as coral reefs and sea grass.



NON-NATIVE SPECIES

Marine debris transports alien and invasive species from one region to another.

Seafood study finds plastic in all samples

A study of five different seafoods has found traces of plastic in every sample tested.



Francisca Ribeiro

Researchers bought oysters, prawns, squid, crabs and sardines from a market in Australia and analysed them using a newly developed method that identifies and measures five different plastic types simultaneously.

The study – by the University of Exeter and the University of Queensland – found plastic levels of 0.04 milligrams (mg) per gram of tissue in squid, 0.07mg in prawns, 0.1mg in oysters, 0.3mg in crabs and 2.9mg in sardines.

"Considering an average serving, a seafood eater could be exposed to approximately 0.7mg of plastic when ingesting an average serving of oysters or squid, and up to 30mg of plastic when eating sardines, respectively," said lead author Francisca Ribeiro, a QUEX Institute PhD student.

polymers

MDPI

Article

Raman Microspectroscopy Detection and Characterisation of Microplastics in Human Breastmilk

Antonio Ragusa [1], Valentina Notarstefano [2,*], Alessandro Svelato [3], Alessia Belloni [2], Giorgia Gioacchini [2], Christine Blondeel [3], Emma Zucchelli [3], Caterina De Luca [3], Sara D'Avino [3], Alessandra Gulotta [4], Oliana Carnevali [2] and Elisabetta Giorgini [2]

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[3] Department of Obstetrics and Gynecology, San Giovanni Calibita Fatebenefratelli Hospital, Isola Tiberina, Via di Ponte Quattro Capi, 39, 00186 Roma, Italy; alessandrosvelato@virgilio.it (A.S.); christineblondeel@yahoo.fr (C.B.); emmazucchelli16@gmail.com (E.Z.); deluca.caterina33@gmail.com (C.D.L.); saradavino@libero.it (S.D.)
[4] Department of Obstetrics and Gynecology, Università degli Studi di Sassari, Viale S. Pietro, 07100 Sassari, Italy; alessandragulotta87@gmail.com
* Correspondence: v.notarstefano@univpm.it; Tel.: +39-0712204723

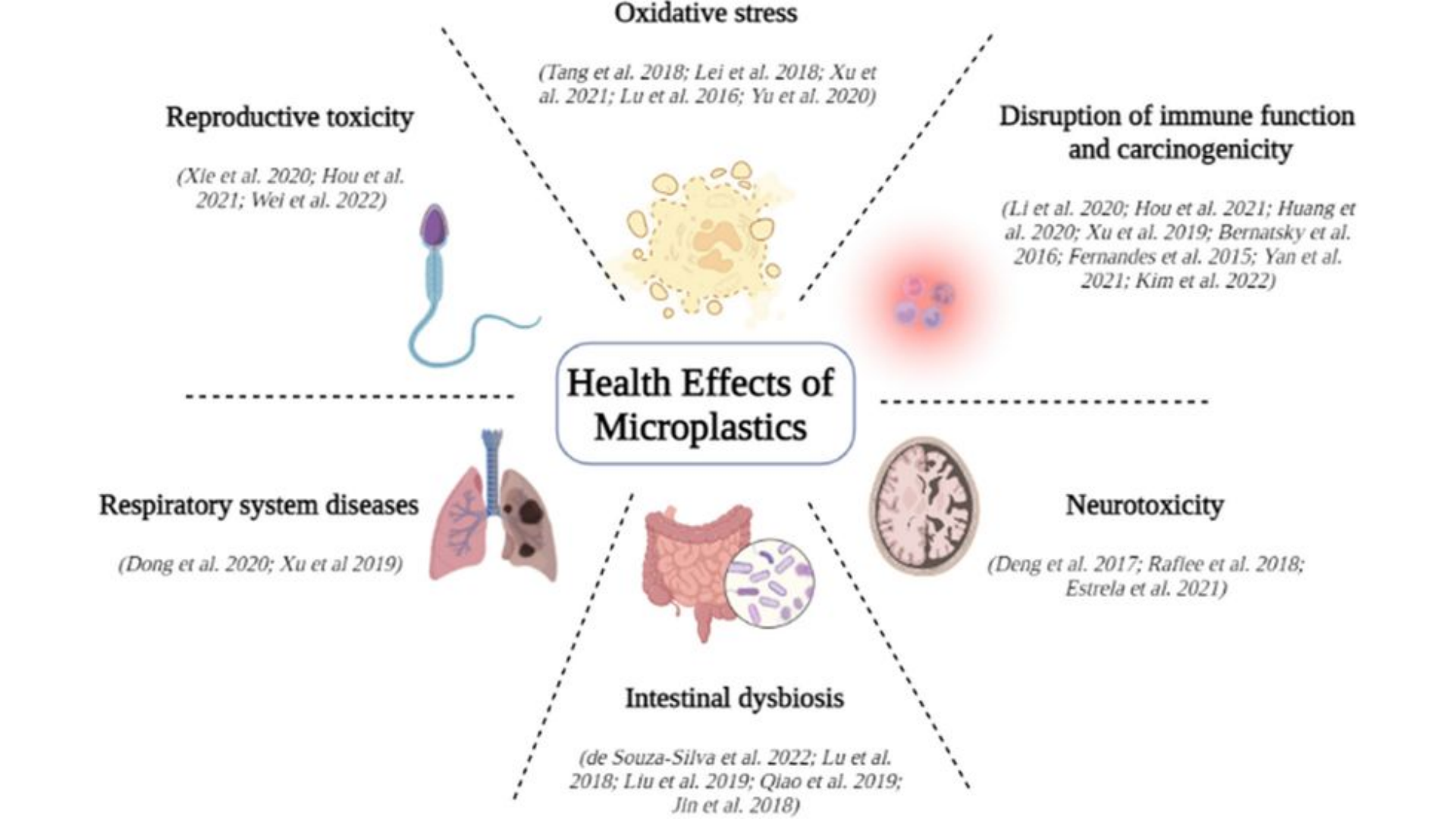
Abstract: The widespread use of plastics determines the inevitable human exposure to its by-products, including microplastics (MPs), which enter the human organism mainly by ingestion, inhalation, and dermal contact. Once internalised, MPs may pass across cell membranes and translocate to different body sites, triggering specific cellular mechanisms. Hence, the potential health impairment caused by the internalisation and accumulation of MPs is of prime concern, as confirmed by numerous studies reporting evident toxic effects in various animal models, marine organisms, and human cell lines. In this pilot single-centre observational prospective study, human breastmilk samples collected from N. 34 women were analysed by Raman Microspectroscopy, and, for the first time, MP contamination was found in 26 out of 34 samples. The detected microparticles were classified according to their shape, colour, dimensions, and chemical composition. The most abundant MPs were composed of polyethylene, polyvinyl chloride, and polypropylene, with sizes ranging from 2 to 12 µm. MP data were statistically analysed in relation to specific patients' data (age, use of personal care products containing plastic compounds, and consumption of fish/shellfish, beverages, and food in plastic packaging), but no significant relationship was found, suggesting that the ubiquitous MP presence makes human exposure inevitable.

check for updates

Citation: Ragusa, A.; Notarstefano, V.; Svelato, A.; Belloni, A.; Gioacchini, G.; Blondeel, C.; Zucchelli, E.; De Luca, C.; D'Avino, S.; Gulotta, A.; et al. Raman Microspectroscopy Detection and Characterisation of Microplastics in Human Breastmilk. *Polymers* **2022**, *14*, 2700. https://doi.org/10.3390/polym14132700

Academic Editors: Xinfeng Wei, Emmanuel Richaud and Andrew M. Booth

Received: 13 June 2022


Oxidative stress
(Tang et al. 2018; Lei et al. 2018; Xu et al. 2021; Lu et al. 2016; Yu et al. 2020)
Reproductive toxicity
(Xie et al. 2020; Hou et al. 2021; Wei et al. 2022)
Disruption of immune function and carcinogenicity
(Li et al. 2020; Hou et al. 2021; Huang et al. 2020; Xu et al. 2019; Bernatsky et al. 2016; Fernandes et al. 2015; Yan et al. 2021; Kim et al. 2022)
Health Effects of Microplastics
Respiratory system diseases
(Dong et al. 2020; Xu et al 2019)
Neurotoxicity
(Deng et al. 2017; Rafiee et al. 2018; Estrela et al. 2021)
Intestinal dysbiosis
(de Souza-Silva et al. 2022; Lu et al. 2018; Liu et al. 2019; Qiao et al. 2019; Jin et al. 2018)

Nature of Science



Scientists can influence the actions of citizens if they provide clear information about their research findings. Popular media coverage of the effects of plastic pollution on marine life changed public perception globally, which has driven measures to address this problem.

D4.2.11—Restoration of natural processes in ecosystems by rewilding

Rewilding

Nature will heal itself if we just leave it alone!!!

Reintroduction of apex predators

Create natural corridors between areas

Minimise human impact!

Hinewai

1987 private nature reserve established.

Left to rewild

Removal of alien species.

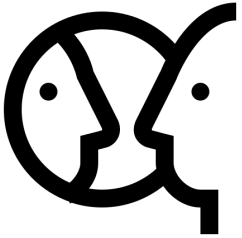






30 minute documentary

Can you...



Explain the requirements for a stable ecosystem?

Calculate percentage change?

Discuss the pros and cons of studies using mesocosms?

Discuss the role of keystone species?

Discuss sustainable harvesting and agriculture?

Explain the process of eutrophication?

Discuss the effects of bioaccumulation and biomagnification?

Discuss the effects of plastic pollution?

Give an example of successful rewilding?